

Muhammad Hassan

AI/ML Engineer

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[LinkedIn](#) • [GitHub](#)

PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in AI/ML engineering with hands-on experience building LLM-powered applications, RAG pipelines, and end-to-end machine learning systems. Proficient in Python, LangChain, ChromaDB, OpenAI API, and Scikit-learn (classification, regression, and clustering algorithms). Developed and deployed multiple real-world ML projects including fraud detection, predictive maintenance, and AI research assistants. Actively seeking an AI/ML internship to apply and expand practical skills.

TECHNICAL SKILLS

AI / LLM: LangChain, OpenAI API, Cerebras API, Exa API, RAG Pipelines, Prompt Engineering, ChromaDB

ML & Data Science: Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Feature Engineering, Data Preprocessing

Programming: Python, C++, SQL, OOP, Data Structures & Algorithms

Tools & Platforms: Git, GitHub, Streamlit, Jupyter Notebook, VS Code

Soft Skills: Analytical Thinking, Problem Solving, Self-Directed Learning, Adaptability, Team Collaboration

Languages: English (Intermediate), Urdu (Native), Punjabi (Native)

PROJECTS

AI Multi-Agent Research Assistant | *Python · LangChain · ChromaDB · Cerebras · Exa API*

- Architected a multi-agent AI research system combining real-time web retrieval (Exa API) and LLM reasoning (Cerebras) to generate comprehensive, source-backed research reports.
- Implemented a two-stage RAG pipeline with AI-driven query expansion and iterative follow-up searches, improving research depth and answer accuracy.
- Engineered autonomous agents to analyze fundamentals, emerging trends, and industry applications, synthesizing findings into actionable executive summaries.

GitHub: [View Code](#)

Credit Card Fraud Detection | *Python · Scikit-learn · SMOTE · Random Forest · Streamlit*

- Built an end-to-end fraud detection system on 284,807 real transactions (0.17% fraud rate), handling extreme class imbalance using SMOTE oversampling and class-weighted Random Forest.
- Trained and compared Naive Bayes, KNN, and Random Forest classifiers, achieving 92% precision, 72.6% recall, and 81.2% F1-score, validated using ROC-AUC curve analysis.
- Deployed a live interactive Streamlit web app supporting real-time single-transaction prediction and batch CSV upload, processing full 284K-row dataset in production.

Live Demo: [Streamlit App](#) • **GitHub:** [View Code](#)

Predictive Maintenance Failure Classification | *Python · Scikit-learn · LDA · Random Forest · Pandas*

- Built a machine failure prediction system on a 10,000-sample industrial dataset, engineering a derived 'Power' feature from torque and rotational speed to improve model accuracy.
- Trained and compared four classification algorithms (LDA, Naive Bayes, KNN, Random Forest), evaluating performance using accuracy, precision, recall, F1-score, and AUC-ROC, with LDA delivering the best results validated through 5-fold cross-validation.
- Identified two distinct failure modes (Power Failure vs Overstrain Failure) via LDA coefficient analysis and deployed the model using Jupyter for real-time failure-risk prediction.

GitHub: [View Code](#)

Heart Disease Prediction | *Python · Scikit-learn · Random Forest · Logistic Regression · Naive Bayes · KNN*

- Trained and compared four classification algorithms (Random Forest, Logistic Regression, Naive Bayes, KNN), with Random Forest achieving 87% accuracy and 86.79% F1-score on patient clinical data.
- Deployed a live Streamlit web app for real-time heart disease risk prediction with probability score and risk level indicator across all 4 models simultaneously.

Live Demo: [Streamlit App](#) • **GitHub:** [View Code](#)

EDUCATION

B.Sc. Computer Science — Major: Software Development

Oct 2023 – Oct 2027

University of Central Punjab, Lahore

Intermediate in Computer Sciences

Nov 2021 – Sep 2023

Punjab Group of Colleges, Lahore

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification — Coursera
- AWS Academy Graduate — Cloud Foundations
- Zero to Hero in LangChain: Build GenAI Apps — Udemy
- Python Programming